

ACHINTYA NIGAM

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Education

Munster Technological University (MTU)

MSc Cybersecurity

Cork, Ireland

Sep 2026 – Present

Vellore Institute of Technology (VIT)

B.Tech Computer Science and Engineering — CGPA: 9.02/10 (First Class with Distinction)

Vellore, India

2022 – 2026

Experience

Independ

Junior Software Developer

Vellore, India

Dec 2024 – Mar 2025

- Engineered Kairos, a React Native client management system, implementing background processing services designed to securely handle and store sensitive customer call data.
- Maintained application architecture and data integrity while optimizing automated, security-conscious data workflows for Android environments.

Independ

Frontend Developer Intern

Vellore, India

Jul 2024 – Nov 2024

- Developed the Next.js frontend for the Independ platform, implementing secure API data consumption and safe user-input handling.

Alumnus Software

Software Engineer Trainee Intern

Remote

Jan 2024 – Jun 2024

- Developed scalable Spring Boot and React.js applications with a strong focus on secure backend architecture.
- Secured REST API endpoints and user sessions by implementing JWT-based authentication and bcrypt password hashing.

Cybersecurity Projects & Research

TCP/IP Network Traffic Analysis and Packet Crafting

Linux, tcpdump, Wireshark, hping3, Netcat

- Crafted and analyzed TCP/IP packets using Wireshark, tcpdump, and hping3, investigating differences in Windows ICMP behaviors and undocumented hping3 APD packet construction behaviors.

Post-Quantum Cryptography on Dual-Core Microcontrollers

C, RP2350A, Raspberry Pi Pico 2 W, Kyber, Dilithium

- Optimised CRYSTALS-Kyber and CRYSTALS-Dilithium cryptographic operations using the dual-core RP2350A architecture, achieving approximately 30% reduction in key-generation time for Kyber.
- Evaluated execution time, SRAM utilisation, power consumption and energy per operation to quantify the performance/energy trade-offs of multicore execution.
- Co-authored the resulting research, published in IEEE Access.

SOC Analyst Pathway

Hack The Box

- Used Splunk Processing Language (SPL) to query event logs, hunt for statistical outliers, and investigate potential network intrusions within enterprise environments.
- Analyzed Windows Sysmon event IDs and conducted foundational reverse engineering of x86 assembly routines to identify and trace malware behavior.

Technical Skills

Programming: Python, C, C++, Java, JavaScript, TypeScript

Cybersecurity: Network Security, TCP/IP, Vulnerability Assessment, Malware Analysis, SIEM, Security Monitoring

Security Tools: Wireshark, tcpdump, hping3, Nmap, Splunk, Sysmon

Web / Backend: React, Next.js, Spring Boot, REST APIs, HTML/CSS

Cloud / Databases: AWS, MySQL

Systems / Tools: Linux, Git

Certifications

AWS Certified Solutions Architect – Associate

2026

AWS Certified Cloud Practitioner

2024